

Adjacent Equal Prime Gaps Are Strongly Suppressed Relative to Random Ordering

Marginal accuracy of a Hardy–Poisson model and an empirical repeat-count correction

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Abstract

This paper studies adjacent equal prime gaps in 1,150 nonoverlapping windows centered at 46 scales from 10^{10} through 10^{100} . Across 262,296,428 adjacent gap pairs, the real sequence contains 2,671,087 equal-gap repeats, whereas an exact random permutation of the same gap multiset has expected count 5,252,242.348. The resulting deficit is 2,581,155.348 repeats, or 49.144% suppression. Every individual window lies below its own shuffle expectation. The suppression weakens from 52.336% near 10^{10} to 43.818% near 10^{100} , and a Poisson offset model with an inverse-log x correction provides the best tested scale description.

Modulo 3 explains a large but incomplete part of the phenomenon. The exactly recoverable forbidden gaps 2, 4, 8, 10 account for at least 33.778% of the total deficit, while the admissible gaps 6, 12, 18, 30 remain suppressed by 15.742%. A corrected mean-matched Hardy–Poisson model nearly reproduces the exact-shuffle repeat expectation and, at an exhaustive hybrid resolution of eight exact gaps plus six residual ranges, predicts the marginal gap distribution with pooled total-variation error 0.564%. Thus its principal failure is sequential independence rather than marginal gap frequency. Finally, we develop a calibrated predictor

$$\hat{R} = H \exp\left(-a - \frac{c}{\ln x}\right),$$

where H is the Hardy–Poisson repeat expectation. The fitted model and two compact constant candidates substantially outperform the rule $H/2$. These formulas are empirical and conjectural, not asymptotic theorems.

Results at a glance

Primary count result	$R/E_{\text{shuf}} = 0.508561$, hence 49.144% suppression.
Consistency	All 1,150 windows satisfy $R < E_{\text{shuf}}$.
Modulo-3 lower bound	Gaps 2, 4, 8, 10 explain at least 33.778% of all missing repeats.
Admissible-gap residual	Gaps 6, 12, 18, 30 remain suppressed by 15.742%.
Hardy–Poisson calibration	Correlation with exact-shuffle expectation 0.99999695; mean absolute relative error 0.513%.
Hybrid marginal accuracy	Total-variation error 0.564%; range WMAPE 0.673%.
Practical predictor	$\hat{R} = H \exp(-0.581233 - 4.038405/\ln x)$.
Clean empirical candidate	$\hat{R} \approx H \exp(-1/(e-1) - 4/\ln x)$.
Euler-based alternative	$\hat{R} \approx H \exp(-\gamma - (4 + \gamma^2/2)/\ln x)$.

1 Introduction

Let p_n denote the n th prime and $g_n = p_{n+1} - p_n$ the corresponding prime gap. The statistic studied here is the number of indices for which two adjacent gaps are equal:

$$R = \sum_{i=1}^{N-1} \mathbf{1}(g_i = g_{i+1}).$$

It counts three-prime patterns of the form

$$p, \quad p + g, \quad p + 2g.$$

The central empirical question is deliberately conditional and count-based:

Given exactly the same gaps and gap frequencies observed in a window, how many adjacent equal-gap repetitions occur in the true prime sequence compared with a random ordering of those gaps?

This formulation separates two distinct objects. The marginal distribution determines how often each gap size appears. Sequential dependence determines how those gap sizes are ordered. A random permutation keeps the first object fixed while removing the second.

The study has four connected goals:

1. quantify the total suppression of adjacent equal gaps relative to unrestricted random ordering;
2. determine how the effect changes across scales from 10^{10} to 10^{100} ;
3. identify how much is explained by modular structure, especially modulo 3;
4. assess whether a corrected Hardy–Poisson model predicts the marginal gap distribution well enough to serve as a baseline for repeat prediction.

The analysis is computational and statistical. It does not claim a proof of an asymptotic law. Its purpose is to isolate robust empirical structure, measure uncertainty, and distinguish marginal-frequency accuracy from ordering failure.

2 Data and estimands

2.1 Window design

The dataset contains 46 scales,

$$10^{10}, 10^{12}, \dots, 10^{100},$$

with 25 windows at each scale, for a total of 1,150 windows. Each window has width 20,000,000; adjacent windows at the same scale are separated and nonoverlapping. Across all windows there are 262,297,578 observed gaps and 262,296,428 adjacent gap pairs.

For a window containing N gaps, let c_g be the count of gap value g . Under a uniformly random permutation of the observed multiset, the exact expected number of adjacent equal gaps is

$$E_{\text{shuf}} = \sum_g \frac{c_g(c_g - 1)}{N}. \quad (1)$$

This expectation uses the actual observed frequencies and changes only their order.

The primary effect measures are

$$D = E_{\text{shuf}} - R, \quad (2)$$

$$\mathcal{S} = 1 - \frac{R}{E_{\text{shuf}}}, \quad (3)$$

$$L = \log\left(\frac{R}{E_{\text{shuf}}}\right). \quad (4)$$

We also report real, expected, and missing repeat rates per 100,000 adjacent gap pairs.

2.2 Pooled and scale-balanced estimands

The pooled ratio

$$\frac{\sum_w R_w}{\sum_w E_w}$$

weights windows by their expected repeat counts. It answers the literal total-count question. A random-effects meta-analysis across the 46 scales gives a complementary scale-balanced estimate, preventing early scales with many more repeat opportunities from dominating every summary.

3 Statistical methodology

Uncertainty and model selection were assessed with the following tools.

- **Fixed-grid bootstrap:** 50,000 resamples of the 25 windows within each of the 46 fixed scales.
- **Hierarchical bootstrap:** 50,000 resamples of scales, followed by windows within each sampled scale.
- **Poisson offset regression:** real repeat count as the response and $\log(E_{\text{shuf}})$ or $\log(E_{\text{HP}})$ as an offset.
- **Dependence-robust inference:** two-way clustered covariance by scale and repeated window position; generalized estimating equations as sensitivity analysis.
- **Model validation:** leave-one-power-out prediction, Poisson deviance, RMSE, MAE, and MAPE.
- **Scale heterogeneity:** random-effects meta-analysis, inverse-power meta-regression, rank trend tests, and robust trend estimates.
- **Composition analysis:** a Kitagawa-style decomposition of the change in the real-to-shuffle ratio into changing group composition and changing within-group ratios.

- **Marginal-distribution accuracy:** total variation, weighted absolute percentage error, Kullback–Leibler discrepancy, Hellinger distance, and calibration residuals.

The count scale is so large that conventional p -values are rarely the main scientific quantity. The analysis therefore emphasizes effect magnitude, out-of-sample prediction, uncertainty intervals, and stability across scales.

4 The total-count phenomenon

4.1 Pooled result

Across all windows,

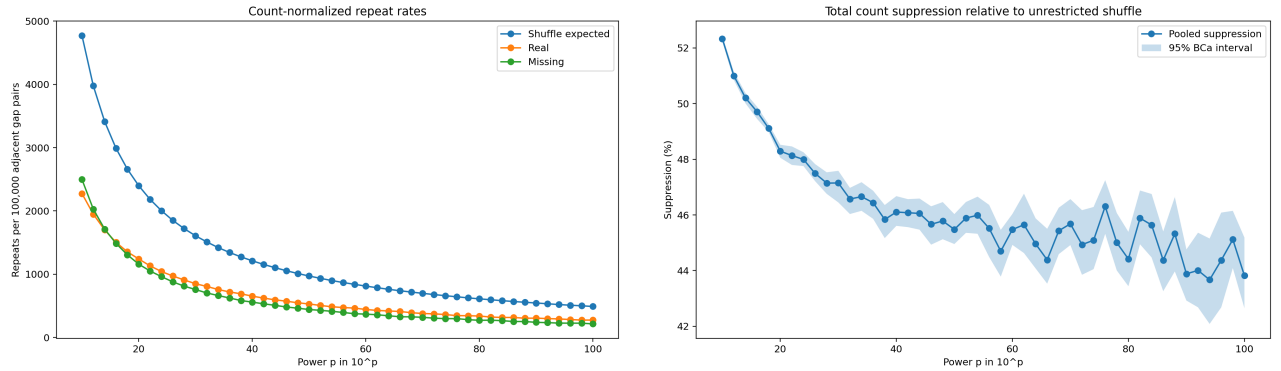
$$\begin{aligned} R &= 2,671,087, \\ E_{\text{shuf}} &= 5,252,242.348, \\ D &= 2,581,155.348. \end{aligned}$$

Thus

$$\frac{R}{E_{\text{shuf}}} = 0.508561, \quad \boxed{\mathcal{S} = 49.144\%}. \quad (5)$$

All 1,150 windows have $R_w < E_w$.

Per 100,000 adjacent pairs, the real sequence has 1018.347 repeats, the shuffle expectation is 2002.407, and the deficit is 984.060. The fixed-grid bootstrap interval for pooled suppression is 49.088%–49.200%. The broader hierarchical interval, which incorporates scale heterogeneity, is 47.323%–50.259%.



(a) Real, expected, and missing repeats per 100,000 adjacent gap pairs. (b) Pooled suppression at each scale with 95% BCa intervals.

Figure 1: The real sequence consistently contains fewer adjacent equal gaps than an unrestricted permutation of the same gap multiset.

4.2 Scale-balanced result

A random-effects analysis that gives the 46 scales balanced influence estimates a real-to-shuffle ratio of 0.537252, corresponding to 46.275% suppression. Its 95% interval for the ratio is $[0.531566, 0.542999]$,

and the prediction interval across scales is $[0.500229, 0.577015]$. The difference from the pooled 49.144% result is expected: lower scales contain far more repeat opportunities and exhibit stronger suppression.

5 Evolution with scale

Suppression decreases from 52.336% near 10^{10} to 43.818% near 10^{100} . The best tested count description uses the shuffle expectation as an offset and an inverse-power correction:

$$\log E[R_w] = \log E_w + \alpha + \frac{\beta}{p_w}, \quad p_w = \log_{10} x_w. \quad (6)$$

The fitted coefficients are

$$\alpha = -0.579769, \quad \beta = -1.628875.$$

The candidate large-scale ratio implied by this fitted form is $e^\alpha = 0.560028$, corresponding to 43.997% suppression. This is a model extrapolation, not a proven limit.

The inverse-power model has Pearson dispersion 0.980 and is the best leave-one-power-out model by Poisson deviance among the tested simple forms. An inverse-power quadratic improves in-sample fit slightly, but the simpler inverse-power description is more stable and interpretable.

5.1 Why suppression weakens

A central question is whether the weakening occurs merely because fixed small forbidden gaps become less common. The data reject that simple explanation.

From 10^{10} to 10^{100} , the real-to-shuffle ratio rises by

$$\Delta(R/E) = 0.085173.$$

The decomposition gives

$$\begin{aligned} \text{composition component} &= -0.016924, \\ \text{within-group component} &= +0.102097. \end{aligned}$$

The within-group component is 119.9% of the net change, while composition offsets part of the increase. Suppression therefore weakens primarily because real-to-shuffle ratios rise inside the gap ranges themselves.

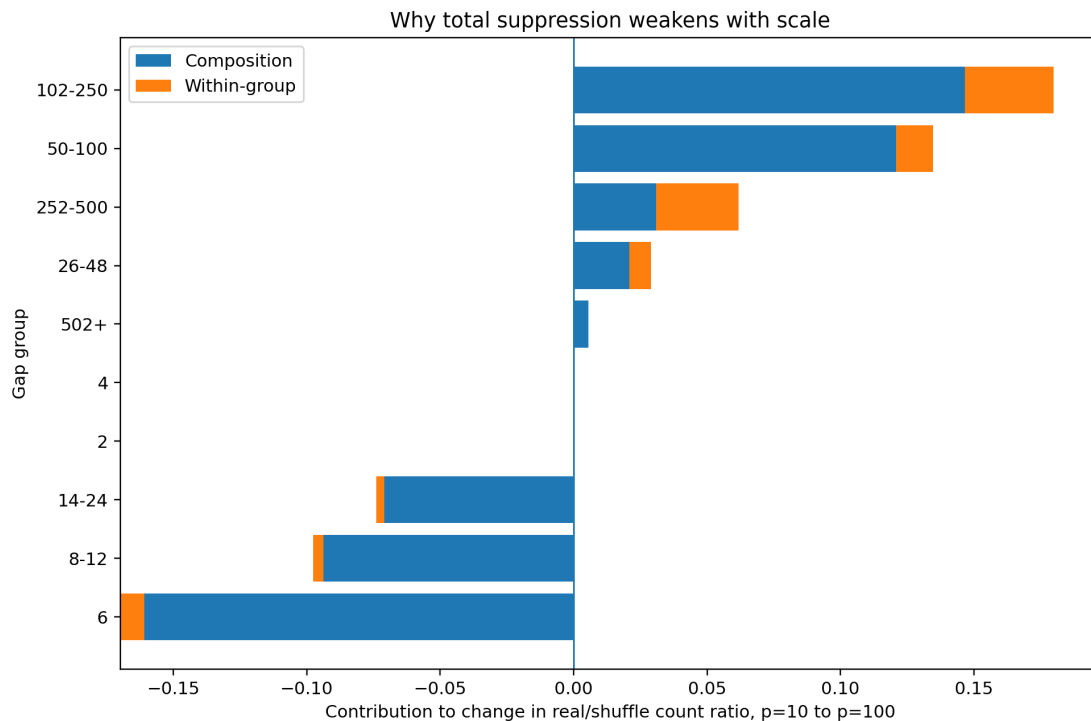


Figure 2: Gap-group contributions to the change in the real-to-shuffle ratio between 10^{10} and 10^{100} . Positive values weaken suppression; negative values strengthen it.

6 Which gap ranges generate the deficit?

Table 1 distinguishes suppression within a group from the group’s contribution to the total missing count. A rare group can be highly suppressed yet contribute little to the global deficit.

Table 1: Pooled gap-group contributions.

Gap group	Real/shuffle ratio	Suppression	Share of deficit
14–24	0.4616	53.845%	24.313%
8–12	0.4537	54.625%	20.395%
26–48	0.5671	43.288%	18.337%
50–100	0.5366	46.337%	10.459%
2	0	100%	8.369%
4	0	100%	8.365%
6	0.8172	18.281%	5.301%
102–250	0.5676	43.242%	3.985%
252–500	0.5751	42.490%	0.448%
502+	0.5340	46.597%	0.029%

The three ranges 8–12, 14–24, and 26–48 account for 63.045% of the total deficit. The phenomenon is therefore not concentrated only in gaps 2 and 4.

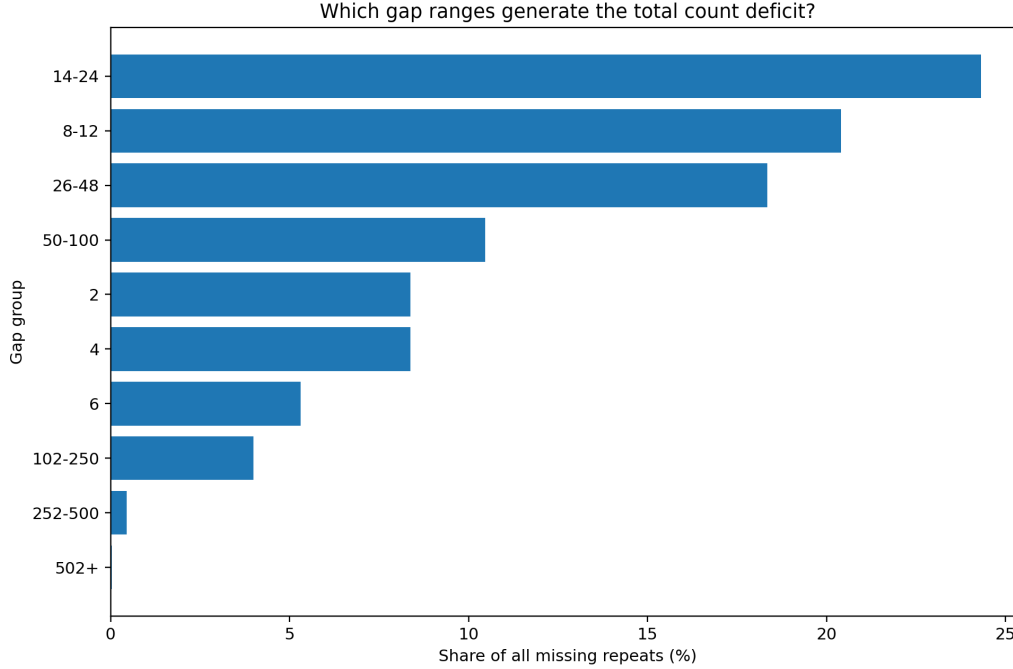


Figure 3: Share of all missing repeats contributed by each exported gap group.

7 Modulo 3 as an explanatory mechanism

The main statistic retains every gap. Modulo 3 is used to explain part of the result, not to redefine the result after removing structurally forbidden events.

Proposition 1 (Repeated-gap divisibility). *For primes larger than 3, an equal adjacent-gap pattern g, g is possible only if $3 \mid g$.*

Proof. The three primes would be $p, p + g, p + 2g$. If $3 \nmid g$, these three numbers occupy all three residue classes modulo 3, so one is divisible by 3. Since the numbers exceed 3, that member cannot be prime. $\square$

The exactly recoverable forbidden gaps 2, 4, 8, 10 contribute 871,861.603 expected-but-impossible repeats. This is at least 33.778% of the total deficit and 16.600% of the entire shuffle expectation. It is a lower bound because other forbidden gaps remain mixed inside aggregate ranges.

The admissible exact gaps 6, 12, 18, 30 show a different fact. Together they have 1,678,531 real repeats against an unrestricted-shuffle expectation of 1,992,126.074, producing

$$\boxed{15.742\% \text{ suppression}}.$$

The individual pooled suppressions are 18.281% for gap 6, 16.514% for gap 12, 17.273% for gap 18, and 7.160% for gap 30.

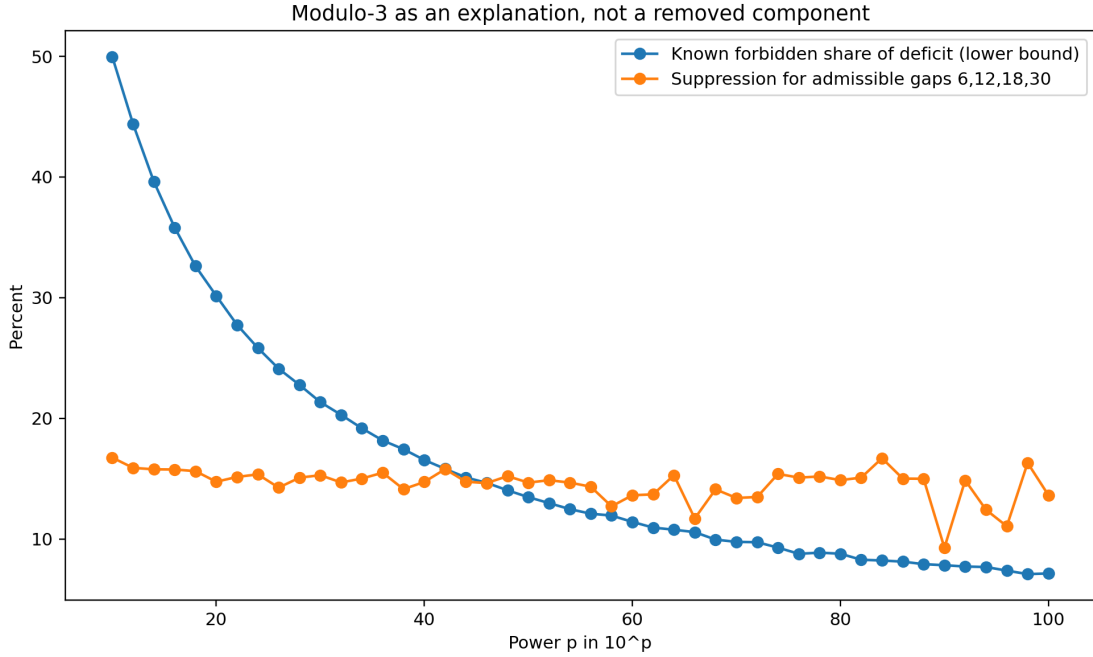


Figure 4: The known forbidden share of the total deficit declines with scale, while selected modulo-3-admissible gaps remain persistently suppressed.

Modulo 3 is therefore a major structural mechanism, but it does not exhaust the observed ordering effect.

8 Exact shuffle and Hardy–Poisson

8.1 Conceptual distinction

The exact shuffle model is empirical and conditional: it takes the actual counts c_g as given and predicts only the repeat count under random ordering via [equation \(1\)](#).

The corrected mean-matched Hardy–Poisson model instead assigns theoretical probabilities p_g to even gaps. Its independent-draw repeat expectation is approximately

$$E_{\text{HP}} \approx (N - 1) \sum_g p_g^2. \quad (7)$$

Thus Hardy–Poisson predicts both the marginal gap distribution and the repeat count implied by independent ordering, whereas shuffle predicts the latter conditional on the observed marginal distribution.

8.2 Hardy–Poisson nearly reproduces shuffle

Across all 1,150 windows, Hardy–Poisson and exact shuffle have Pearson correlation 0.99999695. The mean absolute relative error is 0.513%, relative RMSE is 0.616%, and the largest absolute relative

error is 1.552%. The pooled Hardy–Poisson/shuffle ratio is 1.008992.

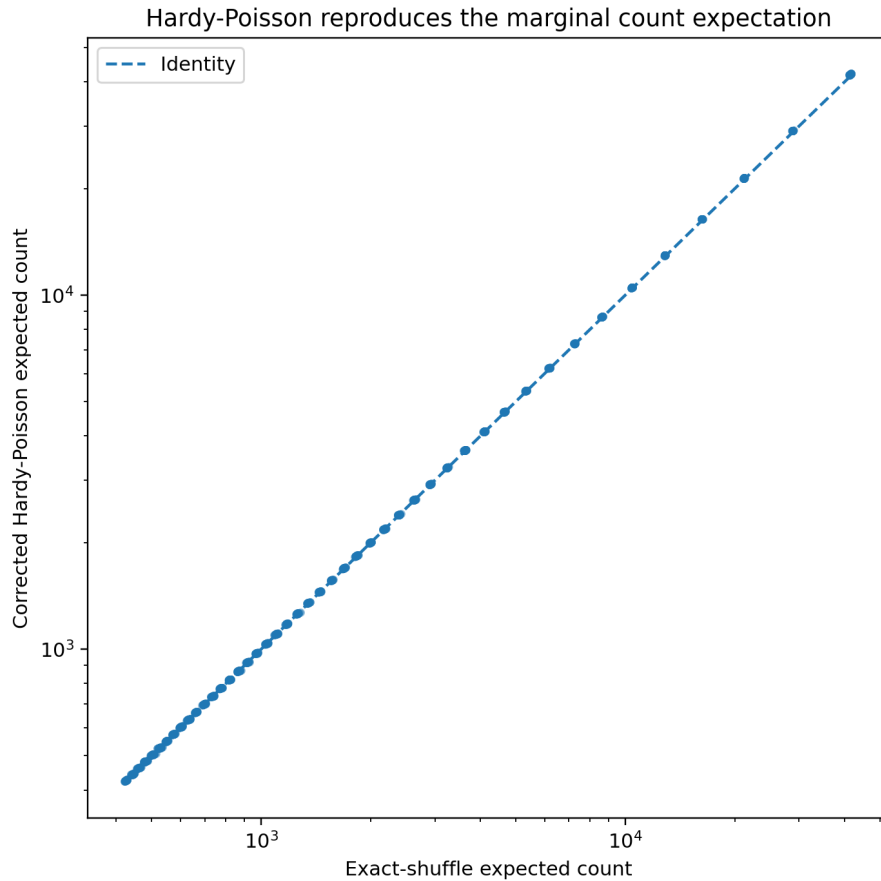


Figure 5: Corrected Hardy–Poisson expected repeat counts against exact-shuffle expected counts at the 46 scales. The near-identity relation suggests that both models encode almost the same marginal collision probability.

This agreement strongly suggests that the corrected Hardy–Poisson model captures the marginal concentration relevant to repeats. It does not by itself prove per-gap accuracy, which motivates the hybrid test below.

9 Hybrid test of marginal gap accuracy

9.1 Exhaustive hybrid partition

The observed distribution is represented by 14 mutually exclusive and exhaustive bins:

- exact gaps 2, 4, 6, 8, 10, 12, 18, 30;
- 14–24 excluding 18;
- 26–48 excluding 30;
- 50–100, 102–250, 252–500, and 502+.

Gap 8 is recovered exactly from the 8–12 aggregate after subtracting exact gaps 10 and 12. The 14 bins sum to the full gap count in every window.

For observed bin probabilities q_b and Hardy–Poisson probabilities p_b , total variation is

$$\text{TV}(p, q) = \frac{1}{2} \sum_b |p_b - q_b|.$$

The pooled result is

$$\boxed{\text{TV} = 0.564\%},$$

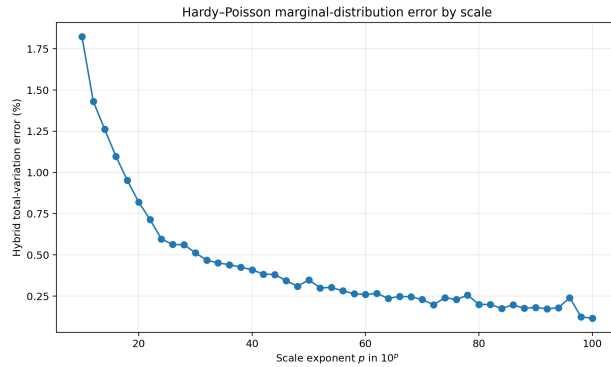
with hierarchical 95% interval 0.376%–0.783%. All-bin WMAPE is 1.128%, exact-gap WMAPE is 2.500%, range WMAPE is 0.673%, and the KL discrepancy is only 0.000230 bits per gap.

The pooled exact-gap calibration is shown in table 2.

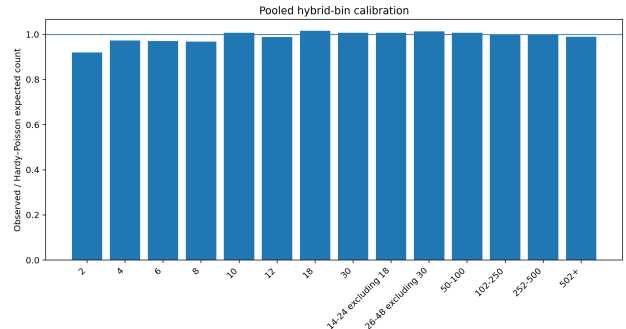
Table 2: Observed exact-gap counts relative to Hardy–Poisson predictions. Positive model bias means Hardy–Poisson predicts too many occurrences.

Gap	Observed/HP	HP bias	Interpretation
2	0.9197	+8.73%	main visible miss
4	0.9728	+2.79%	close
6	0.9709	+3.00%	close
8	0.9679	+3.31%	close
10	1.0073	−0.72%	very close
12	0.9883	+1.19%	very close
18	1.0154	−1.52%	very close
30	1.0070	−0.70%	very close

Every pooled residual range is within approximately 1.33%. The hybrid total-variation error declines from 1.824% at 10^{10} to 0.116% at 10^{100} .



(a) Total-variation error by scale.



(b) Pooled observed/HP ratio for exact gaps and residual ranges.

Figure 6: Hardy–Poisson predicts the marginal gap distribution with high accuracy at the available hybrid resolution.

The interpretation is now stronger than a scalar repeat-count comparison:

Hardy–Poisson accurately predicts how frequently gap sizes occur at the tested hybrid resolution, but it does not predict how those gaps are sequentially arranged.

Broad ranges can conceal within-range cancellation, so the result is not a proof of exact accuracy for every individual gap.

10 A calibrated Hardy–Poisson predictor for real repeats

Let $H = E_{\text{HP}}$ and x be the window center. The crude mental rule

$$\hat{R} \approx H/2$$

is useful, but it ignores the systematic scale trend. A better model is

$$\boxed{\hat{R} = H \exp\left(-a - \frac{c}{\ln x}\right)}. \quad (8)$$

The natural logarithm is used because it avoids an arbitrary log-base dependence in the finite-scale coefficient.

The freely fitted Hardy–Poisson–offset model is

$$\hat{R} = H \exp\left(-0.581233 - \frac{4.038405}{\ln x}\right). \quad (9)$$

The multiplier rises from approximately 0.469 near 10^{10} to approximately 0.549 near 10^{100} . Leave-one-power-out validation gives MAE 31.54 repeats per window, RMSE 50.20, and MAPE 2.900%. By comparison, $H/2$ has MAE 102.85, RMSE 215.73, and MAPE 7.026%.

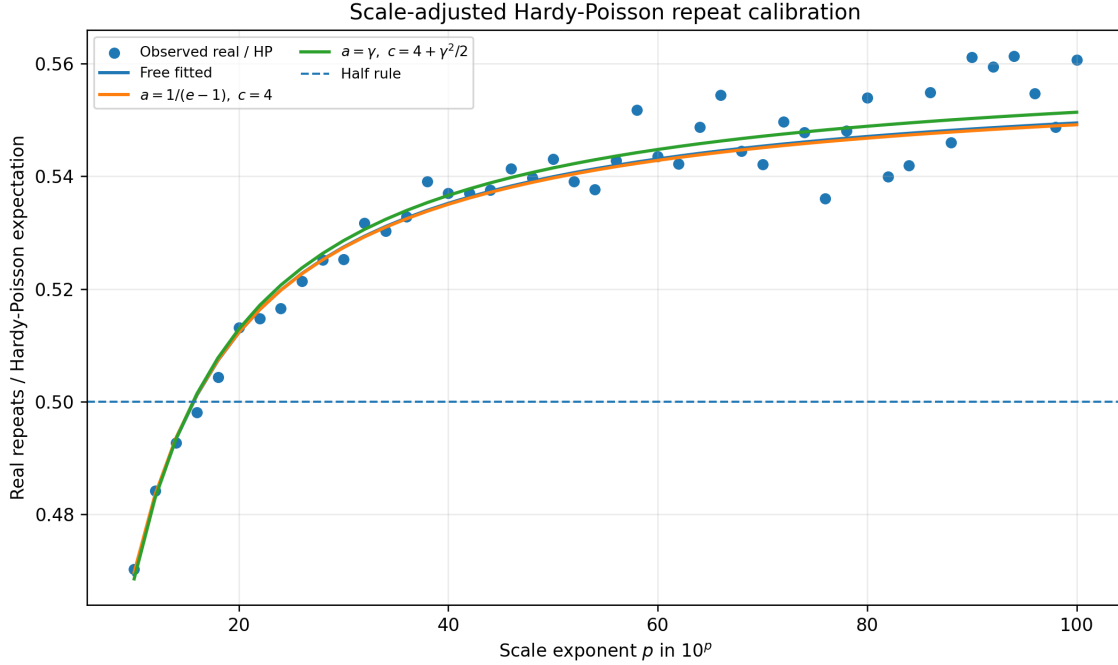


Figure 7: Scale-pooled real/Hardy–Poisson ratios and several calibrated curves. The simple half rule misses the scale dependence.

11 Constant simplifications and conjectural forms

The fitted parameters are empirical. We tested whether low-complexity constants can replace them without sacrificing predictive accuracy.

11.1 Fixing $c = 4$

Fixing the finite-scale coefficient at 4 and refitting the intercept yields

$$a = 0.582209.$$

This one-parameter model slightly improves out-of-sample RMSE through regularization. The strongest clean fully specified candidate found is

$$\hat{R} \approx H \exp\left(-\frac{1}{e-1} - \frac{4}{\ln x}\right). \quad (10)$$

Here

$$\frac{1}{e-1} = 0.5819767069\dots,$$

only 0.0399% from the fitted intercept with $c = 4$. The formula has LOPO MAE 30.91, RMSE 47.97, MAPE 2.895%, and total predicted/actual ratio 1.000232.

11.2 Euler–Mascheroni alternative

If the intercept is fixed at the Euler–Mascheroni constant γ , the fitted finite-scale coefficient is

$$c = 4.165826712\dots$$

A close low-complexity candidate is

$$4 + \frac{\gamma^2}{2} = 4.166588962\dots$$

This gives

$$\hat{R} \approx H \exp\left(-\gamma - \frac{4 + \gamma^2/2}{\ln x}\right). \quad (11)$$

Its LOPO MAE is 31.44, RMSE 49.48, and MAPE 2.901%. The simple rational 25/6 is numerically indistinguishable at the present precision.

Table 3: Leave-one-power-out comparison of repeat-count corrections.

Model	MAE	RMSE	MAPE	Deviance/window
$H \exp(-1/(e-1) - 4/\ln x)$	30.91	47.97	2.895%	1.011
$H \exp(-(7/12) - 4/\ln x)$	30.80	48.22	2.895%	1.014
Fit a , fix $c = 4$	31.02	48.27	2.897%	1.015
$H \exp(-\gamma - (4 + \gamma^2/2)/\ln x)$	31.44	49.48	2.901%	1.018
Free a, c	31.54	50.20	2.900%	1.024
$H/2$	102.85	215.73	7.026%	7.054

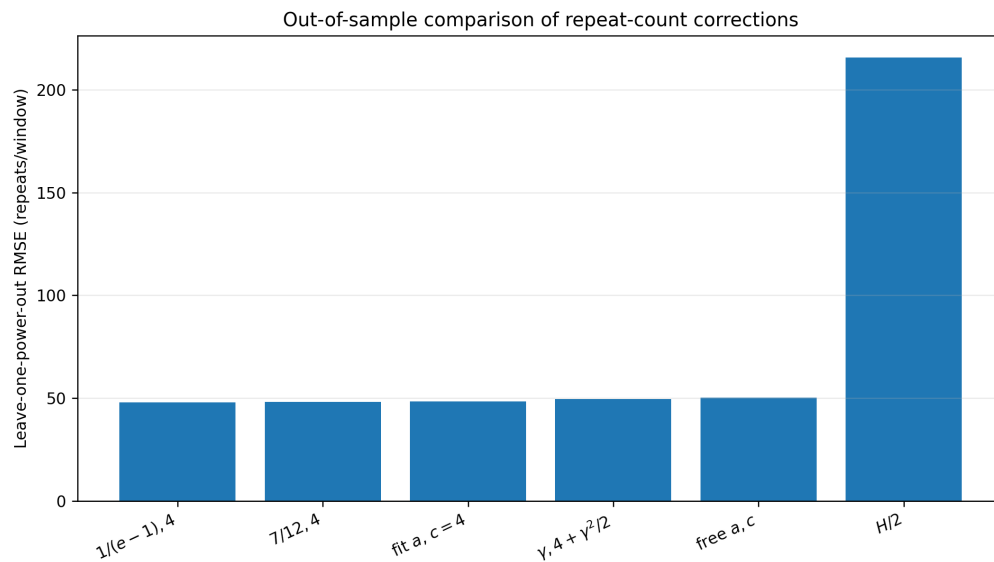


Figure 8: Out-of-sample RMSE for selected corrections. The clean constant candidates are competitive with, and in some metrics slightly better than, the freely fitted two-parameter model.

These constant matches are exploratory and post-selection. The fitted a and c are strongly negatively correlated, so $a \approx \gamma$ and $c \approx 4$ are not independent pieces of evidence. Equations (10) and (11) should be treated as compact empirical conjectures requiring independent confirmation.

12 Interpretation

The results support a coherent two-layer picture.

Marginal layer. The corrected mean-matched Hardy–Poisson model predicts the observed gap distribution extremely well at the tested hybrid resolution. It also nearly reproduces the exact-shuffle collision expectation. Thus the frequency with which gap sizes occur is not the main source of its repeat-count error.

Sequential layer. The true ordering of gaps suppresses adjacent equality by nearly one half relative to unrestricted random permutation. Modulo 3 explains a large exact portion, but admissible gaps remain suppressed, and the weakening with scale happens largely within broad gap ranges. The residual phenomenon therefore concerns structured dependence among neighboring gaps.

The calibrated model [equation \(8\)](#) can be viewed as a two-stage predictor:

1. Hardy–Poisson estimates the marginal gap distribution and its independent repeat expectation H ;
2. the multiplicative correction estimates the ordering-induced suppression of those repeats.

13 Limitations

1. The primary shuffle null is intentionally unrestricted and does not preserve modular residue ordering. It measures the full departure from random ordering.
2. The exact modulo-3 contribution is only partially identifiable from the exported exact-gap data; 33.778% is a lower bound.
3. The hybrid Hardy–Poisson test is not an every-gap calibration. Broad ranges can hide offsetting exact-gap errors.
4. The scale model and constant formulas are empirical calibrations on $10^{10} \leq x \leq 10^{100}$, not proven asymptotic laws.
5. Constant matching was performed after inspecting fitted coefficients and is subject to post-selection optimism.
6. The analysis predicts aggregate repeat counts, not the next individual prime gap or the location of a repeat.
7. Extrapolation outside the studied scale range or to substantially different window designs requires new validation.

14 Conclusion

The real prime-gap sequence contains approximately half as many adjacent equal-gap repetitions as an unrestricted random ordering of the same observed gaps. This statement is not a consequence of marginal gap frequencies alone: the corrected Hardy–Poisson model predicts those frequencies with striking accuracy at hybrid resolution and nearly duplicates the shuffle repeat expectation.

Modulo 3 supplies a substantial structural explanation, accounting for at least one third of the total deficit through four exactly identified forbidden gaps. Yet selected admissible gaps remain suppressed by about 15.7%, and a formal decomposition shows that the weakening of suppression with scale is driven primarily by changes within gap ranges rather than by changing composition alone.

A scale-adjusted multiplicative correction converts the Hardy–Poisson baseline into a useful predictor of real repeat counts. The empirically strongest clean candidate is

$$\hat{R} \approx H \exp\left(-\frac{1}{e-1} - \frac{4}{\ln x}\right),$$

while the Euler-based form

$$\hat{R} \approx H \exp\left(-\gamma - \frac{4 + \gamma^2/2}{\ln x}\right)$$

remains an intriguing alternative. The next mathematical task is not to fit more decimals, but to derive or falsify the structure behind the observed ordering correction.

A Representative scale summaries

Table 4: Selected scale-level pooled results.

Scale	Real count	Shuffle expectation	Ratio	Suppression	Missing/100k
10^{10}	494,015	1,036,445.242	0.476644	52.336%	2498.084
10^{20}	134,682	260,467.716	0.517078	48.292%	1158.455
10^{40}	34,469	64,006.930	0.538526	46.147%	580.890
10^{60}	14,656	26,500.374	0.553049	44.695%	363.671
10^{80}	8,162	14,703.862	0.555097	44.490%	269.520
10^{100}	5,256	9,355.409	0.561816	43.818%	214.689

B Definitions of hybrid accuracy metrics

For observed probabilities q_b and predicted probabilities p_b over the 14 hybrid bins,

$$\begin{aligned}\mathrm{TV}(p, q) &= \frac{1}{2} \sum_b |p_b - q_b|, \\ \mathrm{KL}(q \| p) &= \sum_b q_b \log \frac{q_b}{p_b}, \\ H(p, q) &= \left[\frac{1}{2} \sum_b (\sqrt{p_b} - \sqrt{q_b})^2 \right]^{1/2}.\end{aligned}$$

Weighted absolute percentage error is the absolute count discrepancy summed over bins and divided by the observed total count in the corresponding class of bins.

C Reproducibility of the statistical analysis

The accompanying archive contains the LaTeX source, figures, analysis-ready scale tables, bootstrap intervals, gap-group and modulo-3 summaries, hybrid accuracy tables, constant-model comparisons, a workbook, and the Python script used for the total-count statistical pipeline. The original prime-generation computation is treated as fixed input; the present paper concerns analysis of the resulting data.

References

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